

PROJECT REPORT

Project Title:

Design and Development of Efficient Road object Detection Methods in Extreme Weather conditions for Autonomous Driving.

1. Introduction

Reliable object detection on roads is foundational to the safety and functionality of autonomous vehicles, especially during adverse weather conditions. This report focuses on the inference of Pix2Pix Conditional GAN, with attention-gates at skip connections, trained on Smoke or haze-filled images.

2. Smoke Removal using Pix2Pix GAN (with attention)

Dataset: SMOKE.zip hazy/ and clean/ folders

Link : https://drive.google.com/file/d/1HevtcMrK-GYLW9E_a2sC-kl8enFy13PG/view

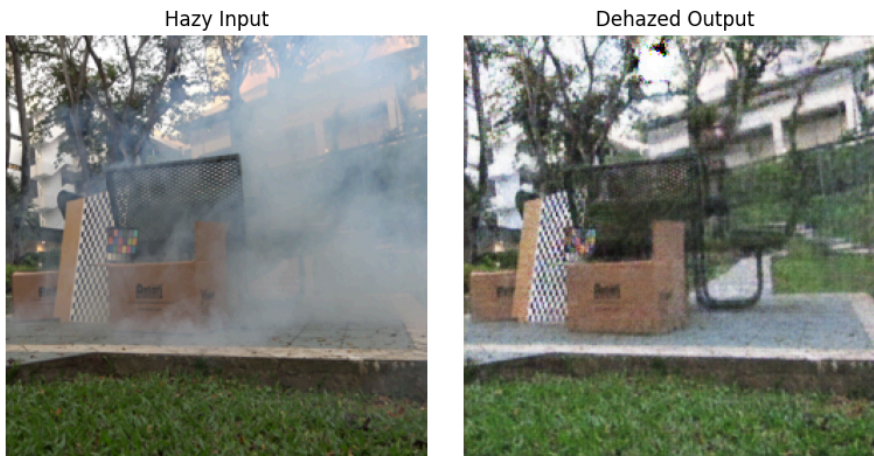
Colab notebook - [here](#)

We have paired a hazy and a corresponding clean image and passed it as input to the generator. - 256x256 pixels.

Model & Training

- Generator - Attention U-Net - 6 downsampling, 5 upsampling, Tanh output, with attention gates around the skip connections
- Discriminator - PatchGAN - 4 convolutional blocks - 1 patch-based output 1 paired input channel
- Training - Loss: **BCELogitsLoss**(GAN) and **L1 loss** (lambda = 100), Optimizer - **Adam** - (learning rate = **1e-4**, beta = 0.5, 0.999), **100 epochs**

3. Output Images:



4. Results

1. Average PSNR - 25.31 dB (signal-to-noise ratio)
2. Average SSIM: 0.7906 (7 percent improvement attributed to inclusion of attention gates)

5. Next steps

Based on the initial results, the project will proceed in three main directions: 1) enhancing the current model's accuracy, 2) performing a rigorous benchmark against other methods, and 3) optimising the model for real-time inference on an embedded system.

We will target to run it on NVIDIA Jetson platform.

- **Model Quantization:** The current FP32 model will be converted to a more efficient, lower-precision format (e.g., FP16 or INT8). This drastically reduces memory footprint and computational load.
- **TensorRT Optimization:** We will use NVIDIA's TensorRT SDK to optimize the model graph. TensorRT performs layer fusion, kernel auto-tuning, and other optimizations to maximize inference throughput on NVIDIA GPUs.

- **Real-Time Performance Testing:** The final optimized model will be deployed on the Jetson, and we will measure its inference speed (frames per second) and power consumption to ensure it meets real-time processing requirements.